INCH-POUND

MIL-DTL-1163B
16 January 2003
SUPERSEDING
MIL-B-1163A
2 September 1988

DETAIL SPECIFICATION

BARS, FLINDERS

Inactive for new design after 26 December 1990

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of soft iron flinders bars. They are intended for use with compasses.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6605

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1916 DoD Preferred Methods for Acceptance of Product.

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DoD adopted are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM INTERNATIONAL

ASTM A 341	-	Standard Test Method for Direct Current Magnetic Properties of Materials using D-C Permeameters and the Ballistic Test Methods.
ASTM A 596	-	Standard Test Method for Direct-Current Magnetic Properties of Materials using the Ballistic Method and Ring Specimens.
ASTM A 773	-	Standard Test Method for dc Magnetic Properties of Materials using Ring and Permeameter Procedures with dc Electronic Hysteresigraphs.
ASTM A 848	-	Standard Specification for Low-Carbon Magnetic Iron.

(Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article testing inspection in accordance with paragraph 4.2.

3.2 Certification. The actual certification of material composition or magnetic property evaluation shall meet the requirements in ASTM A 848 and be included with shipment of the finished flinders bars.

3.3 Material composition. Flinders bars shall be made of the highest grade commercially pure (99.6 percent) soft iron in accordance with ASTM A 848, type 1, and worked in such a manner as to avoid introduction of mechanical strain due to shear or otherwise.

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3.4 Recycled, recovered, or environmentally preferable materials. Unless otherwise specified herein, all materials covered by this specification shall be new. However, recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.5 Magnetic properties. Magnetic testing shall be conducted in accordance with either ASTM A 341, ASTM A 596, ASTM A 773, or by use of a coercimeter.

3.6 Heat treatment. The bars shall be heated slowly and uniformly to a temperature between 1500 and 1600 ºF. Cooling shall be accomplished in the furnace at a rate not to exceed 350 ºF per hour until the temperature reaches 1000 ºF. The cooling rate from that point shall then proceed at a rate not to exceed 100 ºF per hour until the temperature inside the furnace reaches 400 ºF. At that point, the bars shall be removed from the furnace and cooled in still air to approximately ambient temperature. The physical properties of the bars shall meet the requirements specified in ASTM A 848.

3.7 Dimensions. The completed flinders bars shall be 24 inches in length overall and shall be comprised of four sections, each of which shall be cylindrical and 2 inches in diameter throughout. One section shall be 12 inches in length, another section 6 inches in length, and two sections 3 inches in length.

3.8 Workmanship. Workmanship shall be of highest quality and each section shall have an approved black lusterless finish.

3.9 Case. Unless otherwise specified (see 6.2), the flinders bar case shall be rectangular in shape, of hardwood, well seasoned and finished; the top, sides and bottom shall be 1/2 inch thick; the sides of the case shall be dovetailed and secured together by two inside hinges approximately 2 inches from each end; the hinges shall be secured with brass screws; the whole exterior surface of the case shall be varnished; and when the case is closed, the top shall be secured to the bottom by a brass hook and screw eye near each end of the case. The case shall be approximately 26 inches in length, 5-5/8 inches in width and 3-1/4 inches deep, all measurements overall. The inside of the case shall be lined with green felt and shall be divided into two longitudinal sections, into one of which shall fit the flinders bar. In the other section, there shall be four sections of smooth turned hardwood, similar in lengths to the sections of the flinders bar, but of 1-7/8 inches diameter. The grain of these wood spacers shall be approximately parallel to the length of the spacer. The ends of each longitudinal section of the case inside shall have fitted and secured into them a piece of gray felt about 1/2 inch thick to take up any longitudinal movement or thrust of the bar in transport.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection. When a first article inspection is required (see 3.1), the number of units to be provided for inspection shall be specified (see 6.2). Units provided for first article inspection shall be subjected to the examinations of paragraph 4.4.1. The presence of any dimension or bar not within tolerance shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall be performed in accordance with the inspection provisions set forth herein. A sample of bars shall be selected from each lot in accordance with MIL-STD-1916 inspection level I and examined for attributes. The sample bars shall meet the requirements defined in paragraphs 3.7 and 3.8 when subjected to the examinations of paragraph 4.4.1.

4.4 Test methods.

4.4.1 Examination of bars. Each of the sample flinders bars selected in accordance with paragraph 4.3 shall be visually and dimensionally examined to verify compliance with paragraphs 3.7 and 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contact or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activity within the military department or defense agency, or within the military department's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

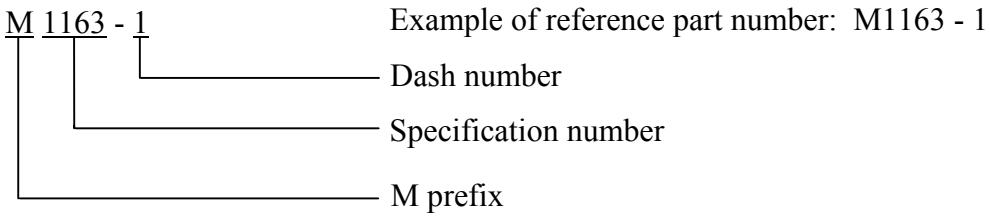
6.1 Intended use. The flinders bars covered by this specification are intended to for use with compasses.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. Whether a first article sample is required (see 3.1).
d. Case, if different (see 3.9).
e. Number of units required for first article inspection (see 4.2).
f. Packaging requirements (see 5.1).

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6.3 Part or identifying number (PIN). The PIN to be used for the flinders bars acquired to this specification is created as follows:



6.4 Subject term (key word) listing.

compass
iron
magnetic
soft

6.5 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy - SH

Preparing Activity:
DLA - GS1

(Project 6605-0521)

Review Activity:
Army - AV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-1163B	**2. DOCUMENT DATE** (YYYYMMDD) 20030116

3. DOCUMENT TITLE
BARS, FLINDERS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center Richmond	b. TELEPHONE (*Include Area Code*) Commercial: (804) 279-5019 DSN: 695-5019
c. ADDRESS *(Include Zip Code)* ATTN: DSCR-VBD (C. Hammond) 8000 Jefferson Davis Highway Richmond, VA 23297-5610	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE STANDARDIZATION PROGRAM OFFICE (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Commercial: (703) 767-6888 DSN: 427-6888

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